

9.2.2 Layout managers

Containers merely keep track of what components they contain. They delegate the positioning and shaping of their components to a layout manager. The `LayoutManager` interface defines methods for laying out components and calculating the preferred and minimum sizes of their containers. The AWT provide five classes that implement either the `LayoutManager` or `LayoutManager2` interfaces.

- `FlowLayout` – specifies that components flow left to right, top to bottom.
- `BorderLayout` – lays out North/South/East/West/Center components.
- `CardLayout` – displays one panel at a time from a deck of panels.
- `GridLayout` – lays out components on a simple grid. Components are stretched to fill the grid.
- `GridBagLayout` – imposes constraints on each component in a grid.

Flow layout

It displays components left to right, top to bottom and is the default layout manager for panels and applets. This just shoves in the components willy-nilly. When an instance of `FlowLayout` is constructed, the alignment (`LEFT`, `CENTER`, `RIGHT`) can be specified, in addition to horizontal and vertical gaps. The horizontal and vertical gaps specify not only the gaps between components but also between the edges of the components and the sides of the container. (This of course is in direct contrast to the `BorderLayout`, where the gaps are only between components.)

Border layout

It lays out components around the sides and in the center of the container in North, East, South, West and Center positions. Gaps between components can be specified. It is the default layout manager for `Window`, `Dialog` and `Frame`. `BorderLayout` is by far the most frequently used layout manager. It is the default manager for most of the AWT containers.

You should use it when a `Container` divides its components into regions: North and South, East and West. Once you've created a few nested layouts, using panels to join components into sub-panels, you'll appreciate the quick convenience of the `BorderLayout`, which sets components (or groups of them) into geographical regions. Nearly every nested layout has a `BorderLayout` lurking somewhere inside.

Here is an example that shows how you can specify the horizontal and vertical gaps between the components: